ASHUTOSH CHANDRAKANT PANDEY

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EDUCATION

Vellore Institute of Technology (VIT), Vellore | 2025–2029
B.Tech, Computer Science and Engineering — Data Science | CGPA: 9.5/10
Class XII: 86.83% | Class X: 86.20% | Branch Rank: 9th

TECHNICAL SKILLS

Programming: Python, C, C++, Java
ML/DL: PyTorch, Scikit-learn, Hugging Face Transformers, BERT, PEFT/LoRA, LIME
Data & Computer Vision: NumPy, Pandas, OpenCV, MediaPipe, Albumentations
Backend & Tools: FastAPI, REST APIs, Docker, Docker Compose, Git, GitHub
Web: HTML5, CSS3, JavaScript

PROJECTS

Unbiasly AI — Bias Detection & Explainability Platform
• Built a bias-detection platform using a BERT-based classifier and parameter-efficient LoRA fine-tuning.
• Integrated LIME explanations to highlight influential text tokens and make predictions easier to interpret.
• Developed FastAPI services and a Docker-based multi-container architecture; added an LLM-assisted text-rephrasing pipeline.

Flood Detection — Semantic Segmentation of Aerial Imagery
• Developed a PyTorch pipeline to segment flooded regions in aerial and drone imagery using U-Net with an EfficientNet-B3 encoder.
• Implemented image resizing, normalization, and augmentation with OpenCV and Albumentations.
• Evaluated segmentation using Dice Score and Intersection over Union (IoU).

Silent Voice — Real-Time Sign Language Interpreter
• Built an ASL gesture-recognition workflow using MediaPipe hand landmarks, OpenCV, and a Scikit-learn MLP classifier.
• Integrated word prediction, multilingual translation, and text-to-speech for spoken output.

TECHNICAL EXPERIENCE & LEADERSHIP

IEEE Computer Society, VIT | AI/ML Domain
• Contributed to the AI/ML hackathon review panel for HackBattle and developed CTF challenges for the What The Flag event.

The AI/ML Club, VIT | Technical Development
• Developed a functional website for the Code Cortex 3.0 event.

Computer Society of India (CSI), VIT | AI/ML Domain
• Contributed to conducting the ForkThis GitHub workshop.

CERTIFICATIONS & ACHIEVEMENTS

• Supervised Machine Learning: Regression and Classification — DeepLearning.AI / Stanford Online
• Advanced Learning Algorithms — DeepLearning.AI / Stanford Online
• Data Science Workshop Certificate — IIT Bombay Techfest
• Academic Achievement: 9th rank in the CSE Data Science branch